# Chapter 2: Random variables and probability distribution

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1. Random variables

2. Probability distribution
   - General concepts
   - PDF and PMF of a discrete random variable
   - pdf of a continuous random variable

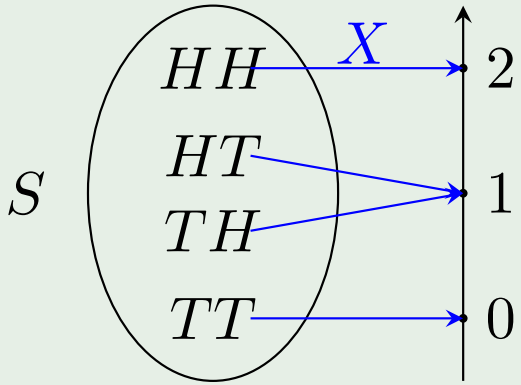
3. Two or more random variables
   - Two discrete random variables
   - Two continuous random variables

# 2.1 Random variables

- Given a random experiment, let S be the corresponding random space.
- A random variable is a mathematical formalization of a quantity which depends on random events

## Example 1

Random experiment: toss two coins and observe the number of heads
Sample space $S = \{HH, HT, TH, TT\}$
$X$: the number of heads



- Random experiment: toss two coins and observe the number of heads
  Sample space $S = \{HH, HT, TH, TT\}$
  $X$: the number of heads
  X can be consider as a function $X : S \longrightarrow \mathbb{R}$

$$X(s) = \begin{cases} 0, & \text{if } s = TT \\ 1, & \text{if } s \in \{HT, TH\} \\ 2, & \text{if } s = HH. \end{cases}$$

- Random experiment: toss a coin and observe the number of tail
  Sample space $S = \{H, T\}$
  $X$: the number of tail

$$X(s) = \begin{cases} 1, & \text{if } s = T, \\ 0, & \text{if } s = H. \end{cases}$$

## Example 2

- Roll a fair die: Let $X$ equal the face number, then $X = 1, 2, 3, 4, 5, 6$.
- Toss a coin until the first heads appears. Let $Y$ be the total number of coin tosses, then $Y = 1, 2, 3, \ldots$
- The random variable $T$ is defined as the time (in hours) from now until the next earthquake occurs in a certain city. The random variable $T$ can in theory get any positive real number in $[0, +\infty)$.

- A random variable is called **discrete** if it has either a finite or a countable number of possible values.
- A random variable is called **continuous** if its possible values contain a whole interval of numbers.
- Denote random variables: $X, Y, Z, \ldots$
- Denote the value of a random variable: $x, y, z, \ldots$

### Example 3

Classify each random variable as either discrete or continuous.

- The number of boys in a randomly selected three-child family.
- The temperature of a cup of coffee served at a restaurant.
- The height of students in a class.
- The number of vehicles owned by a randomly selected household.
- The length of time it takes a truck driver to go from Bien Hoa City to Vung Tau City.

# 2.2 Probability distribution

- Given a random experiment with its associated random variable $X$ and given a real number $x$.
- $P(X \leq x)$ : Probability of the event $\{s \in S \mid X(s) \leq x\}$,
- $P(X < x)$ : Probability of the event $\{s \in S \mid X(s) < x\}$
- $P(X = x)$ : Probability of the event $\{s \in S \mid X(s) = x\}$
- $P(x \leq X \leq y)$ : Probability of the set $\{s \in S \mid x \leq X(s) \leq y\}$ ...

## Definition 4 (Probability distribution function/Cumulative distribution function)

The probability distribution function (PDF), $F_X(x)$, of a random variable $X$ is defined by

$$F_X(x) = P(X \leq x).$$

**Properties:**

- $0 \leq F(x) \leq 1$
- $F(x_1) \leq F(x_2)$ if $x_1 < x_2$
- $\lim_{x \to +\infty} F(x) = F(+\infty) = 1$
- $\lim_{x \to -\infty} F(x) = F(-\infty) = 0$
- $P(a < X \leq b) = F(b) - F(a)$

# PDF and PMF of a discrete random variable

Let $X$ be a discrete random variable assuming values $x_1, x_2, x_3, \ldots$

Probability Distribution Function – PDF

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} P(x_i)$$

*Probability Mass Function – PMF*:

$$P(x) = P(X = x)$$

Properties:

1. $P(x) > 0, \ \forall x \in \{x_1, x_2, \ldots\}$
2. $P(x) = 0$ if $x \notin \{x_1, x_2, \ldots\}$
3. $\sum_{\forall i} P(x_i) = 1$
4. $P(X = x_i) = P(X \leq x_i) - P(X \leq x_{i-1}) = F(x_i) - F(x_{i-1})$
5. $P(a \leq X \leq b) = \sum_{a \leq x_i \leq b} P(x_i)$

## Example 5

Let $X$ be the face number when rolling a single die. Find the probability function for $X$.

**Solution.** Let $X$ be the face number, then $X$ can value $1, 2, 3, 4, 5, 6$ and

$$P(X = 1) = P(X = 2) = P(X = 3) = P(X = 4) = P(X = 5) = P(X = 6) = 1/6$$

| $X$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $P(X = x)$ | 1/6 | 1/6 | 1/6 | 1/6 | 1/6 | 1/6 |

Hence, the pmf is given by

$$p(x) = \frac{1}{6}, \text{ for } x = 1, 2, 3, 4, 5, 6$$

## Example 6

Tossing two fair coins: Let $X$ be the number of heads.

a. Find the probability mass function for $X$.

b. Find the probability distribution function of $X$.

**Solution.** The values of $X$ are 0, 1 or 2. The sample space $S = \{(T,T),(H,T),(T,H),(H,H)\}$. So

$$P(X=0) = 1/4, P(X=1) = 2/4, P(X=2) = 1/4.$$

a. The pmf is given by

$$p(0) = \frac{1}{4}; p(1) = \frac{1}{2}; p(2) = \frac{1}{4}.$$

b. We have

$$F_X(x) = \begin{cases} 0, & x < 0; \\ \frac{1}{4}, & 0 \le x < 1 \\ \frac{3}{4}, & 1 \le x < 2 \\ 1, & x \ge 2. \end{cases}$$

## Example 7

Roll a die repeatedly until getting the first 1. Let $Y$ be the total number of die rolls. Find the distribution mass function of $Y$.

**Solution.** The random variable $Y$ may take any positive integer, i.e $Y = 1, 2, 3, \ldots$. To find the distribution of $Y$, we need to find $P(Y = k)$ for $k = 1, 2, 3, \ldots$. We have

$$P(Y = 1) = \frac{1}{6}; \qquad P(Y = 2) = \frac{5}{6} \times \frac{1}{6}$$

$$P(Y = 3) = \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6}; \qquad P(Y = k) = \left(\frac{5}{6}\right)^{k-1} \times \frac{1}{6}$$

The distribution mass function of $Y$

$$p(k) = \left(\frac{5}{6}\right)^{k-1} \times \frac{1}{6}, \quad k = 1, 2, \ldots$$

## Example 8

I roll a die $n$ times. Let $X$ be the number of occurrences of face 1. Find the distribution mass function of $X$.

**Solution.** If $X = k$, it means that there are $k$ occurrences of face 1 in $n$ rolls. Hence

$$p(k) = \binom{n}{k}\left(\frac{1}{6}\right)^k\left(\frac{5}{6}\right)^{n-k}, \quad k = 0, 1, \ldots, n$$

## Definition 9 (Probability density function)

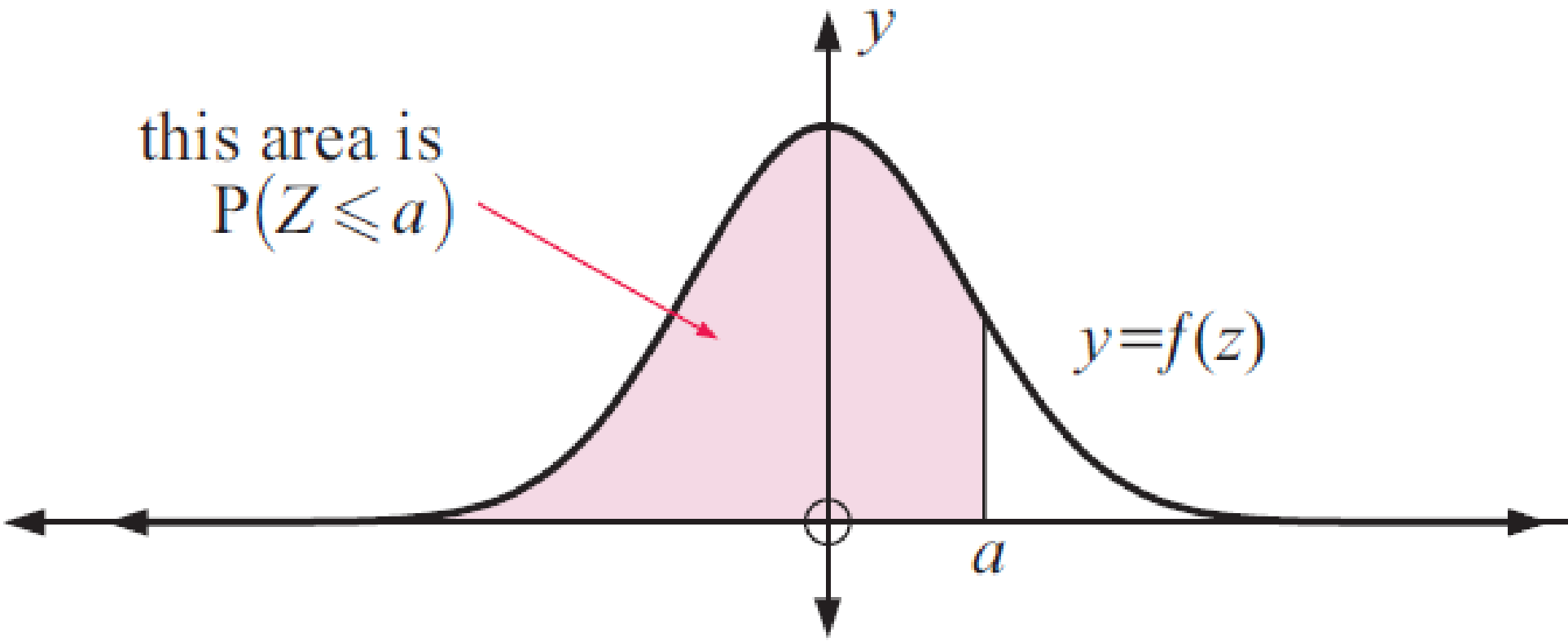
Let $X$ be a continuous random variable. Assume that its PDF, $F_X(x)$, is a continuous function of $x$ and the derivative

$$F'_X(x) := f_X(x)$$

exists for all $x$. The function $f_X(x)$ is called the **probability density function** (pdf), or **density function** of $X$.

Some basic properties of pdf

1. $f_X(x) \geq 0$ for all $x$.
2. $F_X(x) = \int_{-\infty}^{x} f_X(u)\mathrm{d}u$
3. $\int_{-\infty}^{+\infty} f_X(x)\mathrm{d}x = 1$
4. $P(a < X \leq b) = P(a \leq X \leq b) = F_X(b) - F_X(a) = \int_a^b f_X(x)\mathrm{d}x$


y
this area is
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y=f(z)
a

## Example 10

Let the function

$$f_X(x) = \begin{cases} axe^{-x}, & x > 0 \\ 0, & \text{elsewhere;} \end{cases},$$

a. For what value of $a$ is $f$ a pdf?

b. Find $F(x)$.

**Solution.** First note that $f(x) \geq 0$ for all $x$. Now, for $f(x)$ to be a pdf, we need $\int_{-\infty}^{+\infty} f(x)\mathrm{d}x = 1$

$$
\begin{aligned}
1 &= \int_{-\infty}^{+\infty} f(x)\mathrm{d}x \\
&= \int_{0}^{+\infty} axe^{-x}\mathrm{d}x \\
&= a\left[-xe^{-x}\Big|_0^{+\infty} + \int_0^{+\infty} e^{-x}\mathrm{d}x\right] \text{ (using integration by parts)} \\
&= a\left(0 - e^{-x}\Big|_0^{+\infty}\right) = a
\end{aligned}
$$

b. The probability distribution function is

$$
F(x) = \int_{-\infty}^{x} f(t)\mathrm{d}t = \begin{cases} 0, & x \leq 0 \\ \displaystyle\int_0^x te^{-t}\mathrm{d}t = 1 - (x+1)e^{-x}, & x > 0 \end{cases}
$$

## Example 11

Let $X$ be a continuous random variable with density function

$$f(x) = \begin{cases} kx & x \in [0,1] \\ 0 & x \notin [0,1] \end{cases}$$

1. Find all possible value of $k$.
2. Find $F(x)$.
3. Find $P\left(X < \frac{1}{2}\right)$ và $P\left(\frac{1}{2} < X < \frac{3}{4}\right)$.

# Hints

1. $1 = \int_{-\infty}^{+\infty} f(x)dx = \int_{-\infty}^{0} 0dx + \int_{0}^{1} kxdx + \int_{1}^{+\infty} 0dx \qquad \rightarrow k = 2$
2. 

$$F(x) = \int_{-\infty}^{x} f(u)du = \begin{cases} \int_{-\infty}^{x} 0du & \text{if } x \leq 0 \\ \int_{-\infty}^{0} 0du + \int_{0}^{x} 2udu & \text{if } x \in (0,1] \\ \int_{-\infty}^{0} 0du + \int_{0}^{1} 2udu + \int_{1}^{x} 0du & \text{if } x > 1 \end{cases}$$

$$= \begin{cases} 0 & \text{if } x \leq 0 \\ x^2 & \text{if } x \in (0,1] \\ 1 & \text{if } x > 1 \end{cases}$$

3. $P(X < 1/2) = \int_{-\infty}^{1/2} f(x)dx$
   or $P(X < 1/2) = F(1/2)$

   $P(\frac{1}{2} < X < \frac{3}{4}) = \int_{1/2}^{3/4} 2xdx$
   or $P(\frac{1}{2} < X < \frac{3}{4} = F(3/4) - F(1/2)$

## Example 12

The probability density function of a random variable $X$ is given by

$$f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 1, & 1 < x \leq 1.5 \\ 0, & \text{otherwise} \end{cases}$$

a. Find $F(x)$.
b. Find $P(0 \leq X \leq 0.5)$.
c. Find $P(0.5 \leq X \leq 1.2)$.

**Solution.** a. The probability distribution function is

$$F(x) = \int_{-\infty}^{x} f(t)\mathrm{d}t$$

The probability distribution function is

$$F(x) = \begin{cases} 0, & x < 0 \\ \int_0^x f(t)\mathrm{d}t, & 0 \le x < 1 \\ \int_0^1 f(t)\mathrm{d}t + \int_1^x f(t)\mathrm{d}t, & 1 \le x < 1.5 \\ \int_0^1 f(t)\mathrm{d}t + \int_1^{1.5} f(t)\mathrm{d}t, & x \ge 1.5 \end{cases}$$

$$= \begin{cases} 0, & x < 0 \\ \frac{x^2}{2}, & 0 \le x < 1 \\ x - \frac{1}{2}, & 1 \le x < 1.5 \\ 1, & x \ge 1.5 \end{cases}$$

b. $P(0 \le X \le 0.5) = F(0.5) - F(0) = \dfrac{0.5^2}{2} - \dfrac{0^2}{2} = 0.125$

c. $P(0.5 \le X \le 1.2) = F(1.2) - F(0.5) = \left(1.2 - \dfrac{1}{2}\right) - \dfrac{0.5^2}{2} = 0.575$

## Discete random variables

- Probability mas function $P(x)$
  1. $P(x) = P(X = x)$
  2. $P(x) \geq 0 \quad \forall x \in \mathbb{R}$
  3. $\sum_{\forall x} P(x) = 1$
- Probability distribution function

$$F(x) = \sum_{x_i \leq x} P(x_i)$$

- It is NOT always true that $P(X \leq x) = P(X < x)$

## Continuous random variables

- Probability density function $f(x)$:
  1. $f(x) = F'(x) \geq 0 \quad \forall x$
  2. $\int_{-\infty}^{\infty} f(x)dx = 1$
- Probability distribution function

$$F(x) = \int_{-\infty}^{x} f(u)\, du$$

- $P(X = x) = 0 \quad \forall x$
- For every $x$, it is always true that $P(X \leq x) = P(X < x)$

# 2.3 Two or more random variables

## Definition 13

Let $X$ and $Y$ be two random variables.

1. The joint probability distribution function (JPDF) of random variables $X$ and $Y$, denoted by $F(x, y)$, is defined by

$$F_{XY}(x, y) = P(\{X \leq x\} \cap \{Y \leq y\}).$$

2. The marginal distribution function of $X$ (or $Y$)

$$F_X(x) = P(X \leq x) = F_{XY}(x, +\infty);$$

$$F_Y(y) = P(Y \leq y) = F_{XY}(+\infty, y)$$

## Definition 14

The conditional distribution function of a random variable $X$, given that another random variable $Y$ has taken a value $y$, is defined by

$$F_{XY}(x|y) = P(X \leq x|Y = y)$$

**Remark:**
We just use the following formula

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

only if $P(B) \neq 0$.
If $Y$ is a continuous random variable, then $P(Y = y) = 0$ for any value $y$ and so the above formula cannot be applied.

## Definition 15

Let $X$ and $Y$ be two discrete random variables.

1. The joint probability mass function (jpmf) of $X$ and $Y$

$$p(x, y) = P(X = x, Y = y)$$

2. The marginal probability mass functions of $X$ (or $Y$)

$$p_X(x) = \sum_j p(x, y_j); \qquad p_Y(y) = \sum_i p(x_i, y);$$

## Properties

1. $0 \leq p(x, y) \leq 1$
2. $\sum_i \sum_j p(x_i, y_j) = 1$
3. $F(a, b) = \sum_{x \leq a} \sum_{y \leq b} p(x, y)$

## Definition 16

Let $X$ be a discrete random variable. The conditional mass function of $X$ given $Y = y$ is

$$p_{XY}(x|y) = P(X = x|Y = y) \tag{1}$$

Note that

$$p_{XY}(x|y) = P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

or

$$p_{XY}(x|y) = \frac{p_{XY}(x, y)}{p_Y(y)}, \text{ if } p_Y(y) \neq 0.$$

## Definition 17

The discrete random variables $X$ and $Y$ are **independent** if one of the following is hold

- $P(X = x, Y = y) = P(X = x)P(Y = y)$ for all $x, y$;
- $p_{XY}(x, y) = p_X(x) \cdot p_Y(y)$ for all $x, y$.
- $p_{XY}(x|y) = p_X(x)$ for all $x, y$.

**Remark.** To show that $X$ and $Y$ are independent, one usually verifies that that $P(X = x_i, Y = y_j) = P(X = x_i)P(Y = y_j)$, for all $x_i$ and $y_j$.

## Example 18

Consider two random variables $X$ and $Y$ with joint probability mass function given in Table

| | $Y=0$ | $Y=1$ | $Y=2$ |
|---|---|---|---|
| $X=0$ | $\frac{1}{6}$ | $\frac{1}{4}$ | $\frac{1}{8}$ |
| $X=1$ | $\frac{1}{8}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |

a. Find $P(X=0, Y \leq 1)$.

b. Find the marginal PMFs of $X$ and $Y$.

c. Find $P(Y=2|X=1)$.

d. Are $X$ and $Y$ independent?

**Solution.** a. We have

$$P(X=0, Y \leq 1) = p(0,0) + p(0,1) = \frac{1}{6} + \frac{1}{4} = \frac{5}{12}$$

b. Marginal probability mass function of $X$

$$\begin{aligned} p_X(0) &= p(0,0) + p(0,1) + p(0,2) \\ &= \frac{1}{6} + \frac{1}{4} + \frac{1}{8} = \frac{13}{24} \\ p_X(1) &= p(1,0) + p(1,1) + p(1,2) \\ &= \frac{1}{8} + \frac{1}{6} + \frac{1}{6} = \frac{11}{24} \end{aligned}$$

Hence

$$p_X(x) = \begin{cases} \dfrac{13}{24}, & x = 0 \\ \dfrac{11}{24}, & x = 1 \\ 0, & \text{otherwise} \end{cases}$$

Marginal probability mass function of $Y$

$$p_Y(0) = p(0,0) + p(1,0) = \frac{1}{6} + \frac{1}{8} = \frac{7}{24}$$
$$p_Y(1) = p(0,1) + p(1,1) = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$$
$$p_Y(2) = p(0,2) + p(1,2) = \frac{1}{8} + \frac{1}{6} = \frac{7}{24}$$

Hence

$$p_Y(y) = \begin{cases} \frac{7}{24}, & y = 0 \\ \frac{5}{12}, & y = 1 \\ \frac{7}{24}, & y = 2 \\ 0, & \text{otherwise} \end{cases}$$

c. Find $P(Y = 2|X = 1)$.

$$P(Y = 2|X = 1) = \frac{P(Y = 2 \cap X = 1)}{P(X = 1)}$$
$$= \frac{p(1,2)}{p_X(1)} = \frac{1/6}{11/24} = \frac{4}{11}$$

d. We see that

$$P(Y = 2|X = 1) = \frac{4}{11} \neq P(Y = 2) = \frac{7}{24}$$

Therefore $X$ and $Y$ are not independent.

## Definition 19

Let $X$ and $Y$ be continuous random variables. The **joint probability density function** (jpdf), denoted by $f(x, y)$, of $X$ and $Y$

$$f(x, y) := \frac{\partial^2 F(x, y)}{\partial x \partial y}$$

**Properties:**

1. $f(x, y) \geq 0$ for all $x$ and $y$,
2. $F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(u, v) \mathrm{d}u \, \mathrm{d}v$
3. $P(a \leq X \leq b, c \leq Y \leq d) = \int_{c}^{d} \int_{a}^{b} f(x, y) \mathrm{d}x \, \mathrm{d}y$
4. $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) \mathrm{d}x \, \mathrm{d}y = 1.$

- The marginal probability distribution function of $X$

$$F_X(x) = F_{X,Y}(x, +\infty) = \int_{-\infty}^{+\infty} \int_{-\infty}^{x} f(u, v) \mathrm{d}u \mathrm{d}y$$

The marginal probability distribution function of $Y$

$$F_Y(y) = F_{X,Y}(+\infty, y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{y} f(x, v) \mathrm{d}v \mathrm{d}x$$

- The **marginal density functions** of $X$

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) \mathrm{d}y;$$

The **marginal density functions** of $Y$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) \mathrm{d}x.$$

## Definition 20

Let $X$ and $Y$ be continuous random variables.

- The conditional density function of $X$ given $Y = y$ is defined by

$$f_X(x|y) = \frac{\partial F_{XY}(x,y)}{\partial x}$$

If $f_Y(y) \neq 0$ then

$$f_X(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)}$$

**Remark**:

$$P(X \leq a|Y = b) = \int_{-\infty}^{a} f_X(x|b)\mathrm{d}x$$

Two continuous random variables $X, Y$ are **independent** if one of the following is hold

- $f_{XY}(x, y) = f_X(x) \cdot f_Y(y)$ for all $x, y$.
- $f_X(x|y) = f_X(x)$ for all $x, y$.

## Example 21

Let $X$ and $Y$ be two continuous random variables with joint PDF

$$f(x,y) = \begin{cases} x + cy^2, & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

a. Find the constant $c$.
b. Find $P(0 \le X \le \frac{1}{2}, 0 \le Y \le \frac{1}{2})$
c. Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

**Solution.** a. To find $c$, we use

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x,y) \mathrm{d}x \mathrm{d}y = 1$$

Hence

$$1 = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x,y) \mathrm{d}x \mathrm{d}y$$

$$
\begin{aligned}
1 &= \int_0^1 \int_0^1 (x + xy^2)\mathrm{d}x\mathrm{d}y \\
&= \int_0^1 (\frac{1}{2}x^2 + cy^2x)\Big|_0^1 \mathrm{d}y \\
&= \int_0^1 (\frac{1}{2} + cy^2)\mathrm{d}y \\
&= \frac{1}{2}y + \frac{1}{3}cy^3\Big|_0^1 \\
&= \frac{1}{2} + \frac{1}{3}c
\end{aligned}
$$

Suy ra $c = \dfrac{3}{2}$.

b. Find $P(0 \leq X \leq \frac{1}{2}, 0 \leq Y \leq \frac{1}{2})$

$$P(0 \leq X \leq \frac{1}{2}, 0 \leq Y \leq \frac{1}{2}) = \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} (x + \frac{3}{2}y^2)\mathrm{d}x\mathrm{d}y = ... = \frac{3}{32}$$

c. Find the marginal probability density function of $X, Y$. For $0 \leq x \leq 1$,

$$f_X(x) = \int_{-\infty}^{+\infty} f(x,y)\mathrm{d}y = \int_0^1 (x + \frac{3}{2}y^2)\mathrm{d}y$$
$$= (xy + \frac{1}{2}y^3)\Big|_0^1 = x + \frac{1}{2}$$

Therefore,

$$f_X(x) = \begin{cases} x + \frac{1}{2}, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Similarly, for $0 \leq y \leq 1$,

$$f_Y(y) = \int_{-\infty}^{+\infty} f_{XY}(x,y)\mathrm{d}x = \int_0^1 (x + \frac{3}{2}y^2)\mathrm{d}x$$
$$= (\frac{1}{2}x^2 + \frac{3}{2}y^2 x)\Big|_0^1 = \frac{1}{2} + \frac{3}{2}y^2$$

Therefore,

$$f_Y(y) = \begin{cases} \frac{1}{2} + \frac{3}{2}y^2, & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

## Example 22

Let $X$ and $Y$ be two jointly continuous random variables with joint PDF

$$f_{XY}(x,y) = \begin{cases} cx^2y, & 0 \le y \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

a. Find the constant $c$.
b. Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.
c. Find $P(Y \le \frac{X}{2})$.
d. Find $P(Y \le \frac{X}{4} | Y \le \frac{X}{2})$.

**Solution.** a. Find constant $c$.

$$1 = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)\mathrm{d}x\mathrm{d}y$$

$$= \int_0^1\int_0^x cx^2y\mathrm{d}y\mathrm{d}x = \int_0^1 = \int_0^1 \frac{c}{2}x^4\mathrm{d}x = \frac{c}{10}$$

Therefore, $c = 10$.

c. Find the marginal probability density function of $X, Y$. For $0 \leq x \leq 1$,

$$f_X(x) = \int_{-\infty}^{+\infty} f(x,y)\mathrm{d}y = \int_0^x (10x^2y)\mathrm{d}y = 5x^4$$

Therefore,

$$f_X(x) = \begin{cases} 5x^4, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Similarly, for $0 \leq y \leq 1$,

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x,y)\mathrm{d}x = \int_y^1 (10x^2y)\mathrm{d}x = \frac{10}{3}y(1-y^3).$$

Therefore,

$$f_Y(y) = \begin{cases} \frac{10}{3}y(1-y^3), & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

c. Find $P(Y \leq \frac{X}{2})$.

$$P(Y \leq \frac{X}{2}) = \int_{-\infty}^{+\infty} \int_{0}^{\frac{x}{2}} f(x,y)\mathrm{d}y\mathrm{d}x$$
$$= \int_{0}^{1} \int_{0}^{\frac{x}{2}} 10x^2 y\mathrm{d}y\mathrm{d}x$$
$$= \int_{0}^{1} \frac{5}{4}x^4\mathrm{d}x = \frac{1}{4}$$

d. Find $P(Y \leq \frac{X}{4}|Y \leq \frac{X}{2})$. We have

$$P(Y \leq \frac{X}{4}|Y \leq \frac{X}{2}) = \frac{P(Y \leq \frac{X}{4} \cap Y \leq \frac{X}{2})}{P(Y \leq \frac{X}{2})} = \frac{P(Y \leq \frac{X}{4})}{P(Y \leq \frac{X}{2})} = 4P(Y \leq \frac{X}{4})$$
$$= 4\int_{-\infty}^{+\infty} \int_{0}^{\frac{x}{4}} f(x,y)\mathrm{d}y\mathrm{d}x$$
$$= 4\int_{0}^{1} \int_{0}^{\frac{x}{4}} 10x^2 y\mathrm{d}y\mathrm{d}x = 4\int_{0}^{1} \frac{5}{16}x^4\mathrm{d}x = \frac{1}{4}$$

# Homework

T.T.Soong, *Fundamentals of Probability and Statistics for Engineers*, John Wiley & Sons, 2004.
3.1; 3.2; 3.4; 3.5; 3.6; 3.13; 3.16; 3.20